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Curs: Special chapters of differential equations 2025/2026

23/02/2026 - Lecture 1, 02/03/2026 - Lecture 2

Scalar first order differential equations[1]

1. Scalar first order autonomous differential equations

For $f \in C(\mathbb{R})$ we consider the first order scalar autonomous differential equation

$$\dot{x} = f(x). \tag{1}$$

Definition 1 *A solution of* (1) *is a function* $\varphi\colon I \to \mathbb{R}$ *such that* $I \subset \mathbb{R}$ *is an interval with nonempty interior,* $\varphi \in C^1(I)$ *and* $\dot{\varphi}(t) = f(\varphi(t))$ *for all* $t \in I$.

A maximal solution of (1) *is a solution* $\varphi\colon I \to \mathbb{R}$ *such that for any solution* $\psi\colon J \to \mathbb{R}$ *which is an extension of* φ *(i.e.* $I \subset J$ *and* $\psi|_I = \varphi$*) we have* $I = J$ *and* $\psi = \varphi$.

Definition 2 *A point* $\eta \in \mathbb{R}$ *such that* $f(\eta) = 0$ *is called equilibrium point of* $\dot{x} = f(x)$. *Otherwise, it is called regular point.*

The autonomous differential equations have the property that the translation in time of any solution is a solution, too.

Proposition 1 *Let* $\varphi\colon (\alpha, \beta) \to \mathbb{R}$ *be a solution of* (1). *Let* $t_0 \in \mathbb{R}$ *be fixed and consider* $\psi\colon (\alpha + t_0, \beta + t_0) \to \mathbb{R}$, $\psi(t) = \varphi(t - t_0)$. *Then* ψ *is a solution of* (1), *too.*

For each fixed $(t_0, \eta) \in \mathbb{R}^2$ we consider the Initial value problem (IVP), also called Cauchy problem (CP),

$$\dot{x} = f(x), \quad x(t_0) = \eta. \tag{2}$$

Theorem 1 (The existence theorem) *Assume that* $f \in C(\mathbb{R})$. *Then the IVP* (2) *has at least a solution.*

Theorem 2 (The uniqueness theorem) *Assume that* $f \in C^1(\mathbb{R})$ *and that there exist* $\varphi_1\colon I_1 \to \mathbb{R}$ *and* $\varphi_2\colon I_2 \to \mathbb{R}$ *solutions of the IVP* (2). *Then* $t_0 \in I_1 \cap I_2$ *and*

$$\varphi_1 = \varphi_2 \quad on \quad I_1 \cap I_2.$$

Theorem 3 (The existence and uniqueness theorem of the maximal solution) *Assume that $f \in C^1(\mathbb{R})$.*

Then the IVP (2) *has a unique maximal solution, denoted by $\varphi\colon I \to \mathbb{R}$.*

Moreover, $I = (\alpha, \beta)$ is an open interval.

If φ is bounded in the future (i.e. there exists $m > 0$ such that $|\varphi(t)| \leq m$ for all $t \in [t_0, \beta)$) then $\beta = +\infty$.

If φ is bounded in the past (i.e. there exists $m > 0$ such that $|\varphi(t)| \leq m$ for all $t \in (\alpha, t_0]$) then $\alpha = -\infty$.

Note the similarity between the future and the past in the above theorem. In fact, the property for the past can be deduced from the property for the future by considering the equation $\dot{x} = -f(x)$ whose solutions are obtained by reversing the time in the solutions of $\dot{x} = f(x)$. More exactly, if $\varphi\colon (\alpha, \beta) \to \mathbb{R}$ is a solution of $\dot{x} = f(x)$ then $\psi\colon (-\beta, -\alpha) \to \mathbb{R}$, $\psi(t) = \varphi(-t)$, is a solution of $\dot{x} = -f(x)$. So, the past in the equation $\dot{x} = f(x)$ becomes the future in $\dot{x} = -f(x)$.

From now on we assume that

$$f \in C^1(\mathbb{R}).$$

The uniqueness theorem has the following interesting consequences.

Proposition 2 *Assume that $\eta \in \mathbb{R}$ is an equilibrium point of $\dot{x} = f(x)$. Then the unique maximal solution of the IVP* (3) *is the constant function*

$$\varphi\colon \mathbb{R} \to \mathbb{R}, \quad \varphi(t) = \eta.$$

Proposition 3 *Let $\varphi_1\colon I_1 \to \mathbb{R}$ and $\varphi_2\colon I_2 \to \mathbb{R}$ be solutions of $\dot{x} = f(x)$. Assume that there exists $\tau_0 \in I_1 \cap I_2$ such that*

$$\varphi_1(\tau_0) < \varphi_2(\tau_0).$$

Then

$$\varphi_1(t) < \varphi_2(t), \quad t \in I_1 \cap I_2.$$

Proof. Assume, by contradiction, that there exists $\tau_1 \in I_1 \cap I_2$ such that $\varphi_1(\tau_1) \geq \varphi_2(\tau_1)$. Since $\varphi_1(\tau_0) < \varphi_2(\tau_0)$, the continuity of these two functions on the interval $I_1 \cap I_2$ assures that there exists $\tau_2 \in I_1 \cap I_2$ such that $\varphi_1(\tau_2) = \varphi_2(\tau_2)$. Then both functions φ_1 and φ_2 are solutions of the IVP with $\eta = \varphi_1(\tau_2) = \varphi_2(\tau_2)$,

$$\dot{x} = f(x), \quad x(\tau_2) = \eta.$$

The uniqueness theorem assures that $\varphi_1 = \varphi_2$ on $I_1 \cap I_2$. This contradicts the hypothesis. $\square$

Proposition 4 *Let $\varphi\colon I \to \mathbb{R}$ be a solution of $\dot{x} = f(x)$. Assume that there exists $t_0 \in I$ such that $\varphi(t_0)$ is a regular point. Then $\varphi'(t) = f(\varphi(t)) \neq 0$ for all $t \in I$. In particular, φ is strictly monotone.*

Proof. Assume, by contradiction, that there exists $t_1 \in I$ such that $f(\varphi(t_1)) = 0$. Then $\eta = \varphi(t_1)$ is an equilibrium point. We thus have that the IVP $\dot{x} = f(x)$, $x(t_1) = \eta$ is verified by φ and by the constant function η. The uniqueness theorem assures that $\varphi(t) = \eta$ for all $t \in I$. Hence $\varphi(t_0) = \eta$, thus it is an equilibrium point. This contradicts the hypothesis. $\square$

Given a non-constant solution $\varphi\colon (\alpha, \beta) \to \mathbb{R}$ of $\dot{x} = f(x)$ we proved that it is strictly monotone. This implies that there exists $\lim_{t \to \beta^-} \varphi(t) \in \mathbb{R} \cup \{\pm\infty\}$. We will prove in the sequel that, if this limit is finite, then it must be an equilibrium point. We need the following lemma.

Lemma 1 (Barbălat) *Let $\varphi\colon [t_0, \infty) \to \mathbb{R}$ be a C^1 function. If there exist the limits*

$$\lim_{t \to \infty} \varphi(t) \in \mathbb{R} \quad \text{and} \quad \lim_{t \to \infty} \varphi'(t) \in \mathbb{R},$$

then

$$\lim_{t \to \infty} \varphi'(t) = 0.$$

A similar result is valid if $+\infty$ is replaced by $-\infty$.

Proof. Let $m \in \mathbb{N}$. Applying the mean value theorem to the function φ in the interval $[m, m+1]$ we obtain the existence of $\zeta_m \in [m, m+1]$ such that

$$\varphi(m+1) - \varphi(m) = \varphi'(\zeta_m).$$

Since the limit $\lim_{t \to \infty} \varphi(t)$ is finite, we have that $\lim_{m \to \infty} [\varphi(m+1) - \varphi(m)] = 0$. Then $\lim_{m \to \infty} \varphi'(\zeta_m) = 0$. Since $\lim_{m \to \infty} \zeta_m = \infty$ and there exists $\lim_{t \to \infty} \varphi'(t)$ we have that $\lim_{t \to \infty} \varphi'(t) = \lim_{m \to \infty} \varphi'(\zeta_m) = 0$. $\square$

Theorem 4 *Let $\varphi\colon (\alpha, \beta) \to \mathbb{R}$ be a maximal solution of $\dot{x} = f(x)$.*

If $\lim_{t \to \beta^-} \varphi(t) = \eta^* \in \mathbb{R}$ then $\beta = +\infty$ and η^* is an equilibrium point.

If $\lim_{t \to \alpha^+} \varphi(t) = \eta^* \in \mathbb{R}$ then $\alpha = -\infty$ and η^* is an equilibrium point.

Proof. If $\lim_{t \to \beta^-} \varphi(t) = \eta^* \in \mathbb{R}$, then φ is bounded in the future. Theorem 3 assures that $\beta = +\infty$. On the other hand, by the continuity of f we have $\lim_{t \to \infty} f(\varphi(t)) = f(\eta^*)$. Since $\varphi'(t) = f(\varphi(t))$, for all $t \in (\alpha, \infty)$, we obtain that $\lim_{t \to \infty} \varphi'(t) = f(\eta^*)$. The conclusion follows now by the Barbălat's lemma.

The proof of the other case is similar. $\square$

We now apply the above results to the differential equation

$$\dot{x} = 4 - x^2$$

in order to analyze its particular maximal solutions φ and ψ with $\varphi(0) = 0$ and $\psi(0) = 3$.

In this case we have $f(x) = 4 - x^2$. Of course, $f \in C^1(\mathbb{R})$. Thus, indeed, for each $\eta \in \mathbb{R}$, the IVP $\dot{x} = 4 - x^2$, $x(0) = \eta$ has a unique maximal solution.

It is useful to find the equilibrium points solving for $x \in \mathbb{R}$ the equation $f(x) = 0$. They are

$$\eta_1 = -2 \quad \text{and} \quad \eta_2 = 2.$$

Let us denote the maximal solution of the IVP $\dot{x} = 4 - x^2$, $x(0) = 0$ by $\varphi\colon (\alpha, \beta) \to \mathbb{R}$. Since $\varphi(0) = 0 \notin \{-2, 2\}$, we deduce that φ is not a constant function. Thus φ is strictly monotone. Since $\dot{\varphi}(0) = f(\varphi(0)) = f(0) = 4 > 0$, we deduce that

$$\varphi \text{ is strictly increasing.}$$

By Proposition 3, the inequality $-2 < \varphi(0) = 0 < 2$ implies that

$$-2 < \varphi(t) < 2, \quad t \in (\alpha, \beta).$$

Thus, φ is bounded. Theorem 4 assures that

$$(\alpha, \beta) = \mathbb{R}, \quad \lim_{t \to -\infty} \varphi(t) = -2 \quad \text{and} \quad \lim_{t \to \infty} \varphi(t) = 2.$$

Note that the image of φ is $(-2, 2)$, the open interval between the two equilibrium points.

Lets denote the maximal solution of the IVP $\dot{x} = 4 - x^2$, $x(0) = 3$ by $\psi\colon (a, b) \to \mathbb{R}$. Since $\psi(0) = 3 \notin \{-2, 2\}$, we deduce that ψ is not a constant function. Thus ψ is strictly monotone. Since $\dot{\psi}(0) = f(\psi(0)) = f(3) = -5 < 0$, we deduce that

$$\psi \text{ is strictly decreasing.}$$

By Proposition 3, the inequality $2 < \psi(0) = 3$ implies that

$$2 < \psi(t), \quad t \in (a, b).$$

Thus, ψ is bounded in the future. Theorem 4 assures that

$$b = +\infty \quad \text{and} \quad \lim_{t \to \infty} \psi(t) = 2.$$

We have that $\lim_{t \to b^-} \psi(t) > 2$, thus it is not finite since there is no equilibrium point in the interval $(2, \infty)$. Hence,

$$\lim_{t \to b^-} \psi(t) = +\infty$$

and the image of ψ is $(2, \infty)$, an open interval between the equilibrium point 2 and ∞, that do not contain other equilibrium points. $\square$

The next result summarizes the properties of a solution and its image.

Lemma 2 *Let $\varphi\colon (\alpha, \beta) \to \mathbb{R}$ be the maximal solution of the IVP* (2) *and let us denote by γ the image of φ.*

If $f(\eta) > 0$ then φ is strictly increasing and γ is an open interval that do not contain any equilibrium point of $\dot{x} = f(x)$. Moreover, the left end of γ, $\lim\limits_{t\to\alpha^+} \varphi(t)$, is either $-\infty$ or an equilibrium point; while the right end of γ, $\lim\limits_{t\to\beta^-} \varphi(t)$, is either an equilibrium point, or $+\infty$.

A similar result holds when $f(\eta) < 0$.

If $f(\eta) = 0$ then $\gamma = \{\eta\}$.

From now on for each $\eta \in \mathbb{R}$ we consider the IVP with the initial time $t_0 = 0$,

$$\dot{x} = f(x), \quad x(0) = \eta. \tag{3}$$

Denote by $\varphi(\cdot, \eta)\colon I_\eta = \big(\alpha(\eta), \beta(\eta)\big) \to \mathbb{R}$ the unique maximal solution of the IVP (3).

Definition 3 (Dynamical systems vocabulary) *The flow of $\dot{x} = f(x)$ is the function $\varphi\colon D = \{(t, \eta) \in \mathbb{R}^2 \,:\, t \in I_\eta\} \to \mathbb{R}$, $(t, \eta) \mapsto \varphi(t, \eta)$.*

The value $\varphi(t, \eta)$ is said to be the state of the system at time t when starting with the initial state η.

The orbit corresponding to the initial state $\eta \in \mathbb{R}$ is

$$\gamma_\eta = \{\varphi(t, \eta) \,:\, t \in I_\eta\} \subset \mathbb{R}.$$

The phase portrait of $\dot{x} = f(x)$ is the representation on the line $\mathbb{R}$ of the orbits, together with an arrow on each orbit that indicates the future states.

The important properties of the flow are listed below.

Lemma 3 (i) *$\varphi(0, \eta) = \eta$ for any $\eta \in \mathbb{R}$;*

(ii) *$\varphi(t + s, \eta) = \varphi(t, \varphi(s, \eta))$ for any $\eta \in \mathbb{R}$, $s \in I_\eta$, $t \in I_{\varphi(s,\eta)}$;*

(iii) *D is an open subset of $\mathbb{R}^2$ and φ is a C^1 function.*

We have the following **procedure to represent the phase portrait**.

Step 1. Find all the equilibrium points, i.e. solve for $x \in \mathbb{R}$ the equation $f(x) = 0$.

Step 2. Represent the equilibrium points on the line $\mathbb{R}$. By Lemma 2, the orbits are the ones corresponding to each equilibrium point and the open intervals delimited by them.

Step 3. Determine the sign of f on each orbit. According to this sign, insert an arrow on each orbit. If the sign is $+$, the arrow must indicate to the right, while if the sign is $-$, the arrow must indicate to the left.

Definition 4 *An equilibrium solution η^* of $\dot{x} = f(x)$ is said to be an attractor when there is an $r > 0$ such that, whenever $|\eta - \eta^*| < r$ we have $\lim_{t\to\infty} \varphi(t, \eta) = \eta^*$.*

The basin of attraction of an attractor η^* is

$$A_{\eta^*} = \{\eta \in \mathbb{R} \ : \ \lim_{t\to\infty} \varphi(t, \eta) = \eta^*\}.$$

An equilibrium solution η^* of $\dot{x} = f(x)$ is said to be a repeller when there is an $r > 0$ such that, whenever $|\eta - \eta^*| < r$ we have $\lim_{t\to-\infty} \varphi(t, \eta) = \eta^*$.

Theorem 5 (The linearization method) *Let η^* be an equilibrium solution of $\dot{x} = f(x)$. If $f'(\eta^*) < 0$ then η^* is an attractor. If $f'(\eta^*) > 0$ then η^* is a repeller.*

Proof. Assume that $f'(\eta^*) < 0$. Since f is C^1 and $f(\eta^*) = 0$, we have that there is an $r > 0$ such that

$$f(x) > 0, \quad x \in (\eta^* - r, \eta^*) \quad \text{and} \quad f(x) < 0, \quad x \in (\eta^*, \eta^* + r).$$

This is sufficient to conclude that η^* is an attractor. □

Problems.

1. Find the flow $\varphi(t, \eta)$ of $\dot{x} = -2x$. Represent the graphs of the function $\varphi(\cdot, \eta)$ for each $\eta \in \{-2, -1, 0, 1, 2\}$.

2. Find the flow $\varphi(t, \eta)$ of $\dot{x} = x^2$. Represent the graphs of the function $\varphi(\cdot, \eta)$ for each $\eta \in \{-2, -1, 0, 1, 2\}$.

3. We consider $\dot{x} = 4 - x^2$.

(a) Represent the phase portrait. List all the orbits.

(b) For each equilibrium point, decide whether it is an attractor or a repeller.

(c) Reading the phase portrait, list the properties of the maximal solution of the IVP $\dot{x} = 4 - x^2$, $x(0) = \eta$ for each $\eta \in \{-3, -2, -1, 0, 1, 2, 3\}$. Represent the graphs of these functions.

4. We consider $\dot{x} = 3x - x^2$.

(a) Represent the phase portrait. List all the orbits.

(b) For each equilibrium point, decide whether it is an attractor or a repeller.

(c) Reading the phase portrait, list the properties of the maximal solution of the IVP $\dot{x} = 3x - x^2$, $x(0) = \eta$ for each $\eta \in \{-3, -2, -1, 0, 1, 2, 3\}$. Represent the graphs of these functions.

5. We consider $\dot{x} = \sin x$.

(a) Represent the phase portrait. List all the orbits.

(b) For each equilibrium point, decide whether it is an attractor or a repeller.

(c) Reading the phase portrait, list the properties of the maximal solution of the IVP $\dot{x} = \sin x$, $x(0) = \eta$ for each $\eta \in \{-3\pi, -1, 0, 1, \pi/2, 3.14, \pi\}$. Represent the graphs of these functions.

6. Represent the phase portrait of:

(a) $\dot{x} = x - x^3$, (b) $\dot{x} = x - x^3 + 1$, (c) $\dot{x} = x - x^3 + 0.2$, (d) $\dot{x} = -x - x^3 + 1$,
(e) $\dot{x} = 2\sin x$, (f) $\dot{x} = 1 - 2\sin x$, (g) $\dot{x} = 1 - \sin x$, (h) $\dot{x} = 2 - \sin x$.

7. Represent the phase portrait of:

(a) $\dot{x} = -x^3$, (b) $\dot{x} = x^3$, (c) $\dot{x} = x^2$, (d) $\dot{x} = -x^2$.

Note that the only equilibrium point of each of these equations is $\eta^* = 0$ and that Theorem 5 can not be applied for it.

8*. Let $f \in C^1(\mathbb{R})$ and $\eta \in \mathbb{R}$. Let $\varphi\colon (\alpha, \beta) \to \mathbb{R}$ be the maximal solution of

$$\dot{x} = f(x), \quad x(0) = \eta.$$

Assume that $f(\eta) > 0$ and $\lim_{t \to \beta^-} \varphi(t) = +\infty$. Prove that

$$\beta = \int_\eta^\infty \frac{1}{f(u)} du.$$

Proof. We have

$$\dot{\varphi}(t) = f(\varphi(t)), \quad t \in (\alpha, \beta).$$

Since $f(\eta) > 0$, by Proposition 4 we have that

$$f(\varphi(t)) > 0, \quad t \in (\alpha, \beta).$$

Then

$$\frac{\dot{\varphi}(t)}{f(\varphi(t))} = 1, \quad t \in (\alpha, \beta).$$

Integrating from 0 to β with respect to the variable t we obtain

$$\int_0^\beta \frac{\dot{\varphi}(t)}{f(\varphi(t))} dt = \beta.$$

In this integral we make the change of variable $u = \varphi(t)$, which is a good one since φ is injective. Note that $\varphi(0) = \eta$, $\lim_{t \to \beta^-} \varphi(t) = \infty$. $\square$

9. Let $f(x) = x^{1/3}$. Note that $f \in C(\mathbb{R})$, but $f \notin C^1(\mathbb{R})$.
Let

$$\varphi(t) = \begin{cases} 0, & t < 0 \\ \left(\frac{2}{3}t\right)^{3/2}, & t \geq 0\,. \end{cases}$$

Check that φ is a maximal solution of $\dot{x} = f(x)$. Note that the IVP

$$\dot{x} = x^{1/3}, \quad x(0) = 0$$

has at least two maximal solutions.

10. Let $0 < c < 1$ be a parameter and consider the scalar dynamical system $\dot{x} = x(1-x) - cx$.

a) Find its equilibria and study their stability using the linearization method.

b) Represent its phase portrait.

c) When $x(t) > 0$ is considered to be the density of fish in a lake, and $0 < c < 1$ to be the rate of fishing, try to predict the fate of the fish from the lake interpreting the theoretical result obtained at a) and b). $\diamond$

11. Let $c > 1/4$ be a parameter and consider the scalar dynamical system $\dot{x} = x(1-x) - c$.

a) Represent its phase portrait.

b) When $x(t) \geq 0$ is considered to be the density of fish in a lake, and $c > 1/4$ to be the rate of fishing, try to predict the fate of the fish from the lake interpreting the theoretical result obtained at a). $\diamond$

12. Represent the phase portrait of $\dot{x} = \lambda - x^2$. Discuss with respect to the parameter $\lambda \in \mathbb{R}$. $\diamond$

13. Denote by $x(t)$ the density of a trout population at time t. Describe its evolution in each of the following cases.

(a) $\dot{x} = 100x$.

(b) $\dot{x} = 100x - x^2$.

Assume that someone is fishing, first with a constant rate $k > 0$, that is,

(c) $\dot{x} = 100x - x^2 - k$,

and another one, in another basin, with rate proportional to the density, $kx(t)$, where $k > 0$, that is,

(d) $\dot{x} = 100x - x^2 - kx$.

Discuss with respect to k. $\diamond$

14. For each $k > 0$ we consider the differential equation

$$\dot{x} = -k\,(x - 21),$$

which is the model of Newton for cooling processes, here $x(t)$ being the temperature of a cup of tea at time t.

(a) Find its flow. (b) An experiment revealed the following fact. A cup of tea with initial temperature of $49°C$ has a temperature of $37°C$ after 10 minutes. Find the initial temperature of a cup of tea such that after 20 minutes the tea has $37°C$. $\diamond$

2. Scalar first order differential equations

For $\Omega \subset \mathbb{R}^2$ nonempty, open and connected, and for $f \in C(\Omega)$ we consider the first order scalar autonomous differential equation

$$x' = f(t, x). \tag{4}$$

Definition 5 *A solution in Ω of (4) is a function $\varphi\colon I \to \mathbb{R}$ such that $I \subset \mathbb{R}$ is an interval with nonempty interior, $\varphi \in C^1(I)$, $(t, \varphi(t)) \in \Omega$ and $\varphi'(t) = f(t, \varphi(t))$ for all $t \in I$.*

A maximal solution of (4) is a solution $\varphi\colon I \to \mathbb{R}$ such that for any solution $\psi\colon J \to \mathbb{R}$ which is an extension of φ (i.e. $I \subset J$ and $\psi|_I = \varphi$) we have $I = J$ and $\psi = \varphi$.

For each fixed $(t_0, \eta) \in \Omega$ we consider the Initial value problem (IVP), also called Cauchy problem (CP),

$$x' = f(t, x), \quad x(t_0) = \eta. \tag{5}$$

Theorem 6 (The existence theorem) *Assume that $f \in C(\Omega)$. Then the IVP (5) has at least a solution.*

Theorem 7 (The uniqueness theorem) *Assume that $f \in C^1(\Omega)$ and that there exist $\varphi_1\colon I_1 \to \mathbb{R}$ and $\varphi_2\colon I_2 \to \mathbb{R}$ solutions of the IVP (5). Then $t_0 \in I_1 \cap I_2$ and*

$$\varphi_1 = \varphi_2 \quad on \quad I_1 \cap I_2.$$

Theorem 8 (The existence and uniqueness theorem of the maximal solution) *Assume that $f \in C^1(\Omega)$.*

Then the IVP (5) has a unique maximal solution, denoted by $\varphi\colon I \to \mathbb{R}$.

Moreover, $I = (\alpha, \beta)$ is an open interval.

If $\beta < +\infty$ then, either $\lim\limits_{t \to \beta^-} |\varphi(t)| = \infty$ or there exists a sequence of numbers (t_k) such that $\lim\limits_{k \to \infty} (t_k, \varphi(t_k)) = (\beta, l) \in \partial\Omega$.

If $\alpha > -\infty$ then, either $\lim\limits_{t \to \alpha^+} |\varphi(t)| = \infty$ or there exists a sequence of numbers (t_k) such that $\lim\limits_{k \to \infty} (t_k, \varphi(t_k)) = (\alpha, l) \in \partial\Omega$.

Theorem 9 (Equations with global solutions) *Assume that* $\Omega = (\alpha, \beta) \times \mathbb{R}$ *and let* $f \in C(\Omega)$. *Assume that there exist two functions* $a, b \in C(\alpha, \beta)$ *such that*

$$|f(t,x)| \leq a(t)|x| + b(t), \quad t \in (\alpha, \beta), \quad x \in \mathbb{R}.$$

Then any maximal solution in Ω *of* (4) *is defined on* (α, β).

Problems.

1. Let $\varphi_1 \colon \mathbb{R} \to \mathbb{R}$, $\varphi_1(t) = t$. Justify that φ_1 is not a solution of the differential equation (a) $x' = \frac{x}{t}$, (b) $x' = 1 + \sqrt{t^2 - x^2}$. What about $\varphi_2 \colon (0, \infty) \to \mathbb{R}$, $\varphi_2(t) = t$?

2. Justify that any solution in the first quadrant $\Omega = (0, \infty) \times (0, \infty)$ of $x' = -\frac{t}{x}$ is strictly increasing. Find the expression of the maximal solution in $\Omega = (0, \infty) \times (0, \infty)$ of the IVP $x' = -\frac{t}{x}$, $x(1) = 1$, and represent its graph.

3. Justify that any solution of (a) $x' = \frac{1}{1+t^2+x^2}$, (b) $x' = \frac{1}{t-2x}$ is injective.

4. Justify that the maximal solution in $\Omega = \mathbb{R} \times (0, \infty)$ of $x' = -\frac{t}{x}$, $x(0) = \eta > 0$ has a maximum point in $t = 0$. Find the expression of the maximal solution in $\Omega = \mathbb{R} \times (0, \infty)$ of the IVP $x' = -\frac{t}{x}$, $x(0) = \sqrt{2}$, and represent its graph.

5. Let $f \in C(\Omega)$. Let $\varphi_1 \colon [\alpha_1, \alpha_2] \to \mathbb{R}$ and $\varphi_2 \colon [\alpha_2, \alpha_3] \to \mathbb{R}$ be two solutions of (4). Let $\varphi \colon [\alpha_1, \alpha_3] \to \mathbb{R}$,

$$\varphi(t) = \begin{cases} \varphi_1(t), & t \in [\alpha_1, \alpha_2] \\ \varphi_2(t), & t \in (\alpha_2, \alpha_3]. \end{cases}$$

Prove that, if $\varphi_1(\alpha_2) = \varphi_2(\alpha_2)$, then φ is a solution of (4).

6. Let $f \in C^1(\Omega)$. Let $\varphi_1 \colon I_1 \to \mathbb{R}$ and $\varphi_2 \colon I_2 \to \mathbb{R}$ be solutions of $x' = f(t, x)$. Assume that there exists $\tau_0 \in I_1 \cap I_2$ such that

$$\varphi_1(\tau_0) < \varphi_2(\tau_0).$$

Prove that

$$\varphi_1(t) < \varphi_2(t), \quad t \in I_1 \cap I_2.$$

7. Justify that the maximal solution of the IVP

$$x' = (4 - x^2)e^{2tx}, \quad x(0) = 0$$

is bounded, strictly increasing, defined on $\mathbb{R}$.

8. Justify that the IVP

$$x' = x^{1/3}, \quad x(0) = \eta > 0$$

has a unique maximal solution in $\Omega = \mathbb{R} \times (0, \infty)$. Justify that it is strictly increasing and find its interval of definition.

9. Justify that any maximal solution of $x' = -3t \sin(tx) - e^{5t}$ is defined on $\mathbb{R}$.

10. Justify that any maximal solution of $x' = -\frac{3t}{\ln(t-1)} \sin(tx) - e^{5tx^2}$ is defined either on $(1, 2)$ or $(2, \infty)$.

11. Reading the phase portrait of $x' = \frac{1}{3-\cos x}$, note that any maximal solution of it is unbounded and strictly increasing. Prove in two ways that any maximal solution is defined on $\mathbb{R}$. What can be said about the exact (or closed-form) expression of the maximal solution of the IVP $x' = \frac{1}{3-\cos x}$, $x(0) = 0$?

12. Let $J_1, J_2 \subset \mathbb{R}$ be open intervals and $a \in C(J_1)$, $g \in C^1(J_2)$. Prove that for any nonconstant maximal solution $\phi : I \to \mathbb{R}$ of $x' = a(t)g(x)$ we have $g(\phi(t)) \neq 0$ for all $t \in I$.

13. Let $f \in C^2(\mathbb{R}^2)$ and $t_0 \in \mathbb{R}$.
For each $\eta \in \mathbb{R}$ denote by $\varphi(\cdot, \eta)\colon I_\eta \to \mathbb{R}$ the maximal solution of the IVP

$$x' = f(t,x), \quad x(t_0) = \eta.$$

It is known that φ is a C^2 function on $E = \{(t,\eta) \in \mathbb{R}^2 \ :\ t \in I_\eta\}$.
Moreover, if f is C^n then φ is C^n.
Also, for each $k \in \mathbb{N}^*$ denote by $\psi_k\colon I_k \to \mathbb{R}$ the maximal solution of the IVP

$$x' = f(t,x), \quad x(t_0) = \eta_k,$$

where $(\eta_k)_{k\geq 1}$ is a given sequence of real numbers such that there exists $\lim\limits_{k\to\infty} \eta_k = \eta^* \in \mathbb{R}$.
(a) Justify that $\lim\limits_{k\to\infty} \psi_k(t) = \varphi(t, \eta^*)$.
(b) Let

$$u(t) = \frac{\partial \varphi}{\partial \eta}(t, \eta^*), \quad a(t) = \frac{\partial f}{\partial x}(t, \varphi(t,\eta^*)).$$

Justify that

$$u'(t) = a(t)u(t), \quad u(t_0) = 1.$$

14. Let $f \in C^2(\mathbb{R}^3)$, $t_0 \in \mathbb{R}$ and $\eta \in \mathbb{R}$.
For each $\lambda \in \mathbb{R}$ denote by $\varphi(\cdot, \lambda)\colon I_\lambda \to \mathbb{R}$ the maximal solution of the IVP

$$x' = f(t,x,\lambda), \quad x(t_0) = \eta.$$

It is known that φ is a C^2 function on $E = \{(t,\lambda) \in \mathbb{R}^2 \ :\ t \in I_\lambda\}$.
Moreover, if f is C^n then φ is C^n.
Also, for each $k \in \mathbb{N}^*$ denote by $\psi_k\colon I_k \to \mathbb{R}$ the maximal solution of the IVP

$$x' = f(t,x,\lambda_k), \quad x(t_0) = \eta,$$

where $(\lambda_k)_{k\geq 1}$ is a given sequence of real numbers such that there exists $\lim\limits_{k\to\infty} \lambda_k = \lambda^* \in \mathbb{R}$.
(a) Justify that $\lim\limits_{k\to\infty} \psi_k(t) = \varphi(t, \lambda^*)$.
(b) Let

$$v(t) = \frac{\partial \varphi}{\partial \lambda}(t, \lambda^*), \quad a(t) = \frac{\partial f}{\partial x}(t, \varphi(t,\lambda^*), \lambda^*), \quad b(t) = \frac{\partial f}{\partial \lambda}(t, \varphi(t,\lambda^*), \lambda^*).$$

Justify that

$$v'(t) = a(t)v(t) + b(t), \quad v(t_0) = 0.$$

15. For each $\eta \in \mathbb{R}$ denote by $\varphi(\cdot, \eta)$ the maximal solution of the IVP

$$x' = x - x^2 + tx^3, \quad x(4) = \eta.$$

Also, for each $k \in \mathbb{N}^*$ denote by ψ_k the maximal solution of the IVP

$$x' = x - x^2 + tx^3, \quad x(4) = \frac{2k+5}{k^2 - 3k + 7}.$$

It is known that φ is a C^∞ function. Note that $\varphi(t, 0) = 0$ for all $t \in \mathbb{R}$. Find

$$\lim_{k\to\infty} \psi_k(t), \quad \varphi(t, 0) \text{ and } \frac{\partial \varphi}{\partial \eta}(t, 0).$$

16. For each $\eta \in \mathbb{R}$ and $\lambda \in \mathbb{R}$ denote by $\varphi(\cdot, \eta, \lambda)$ the maximal solution of the IVP

$$x' = \lambda + 3x + x^2, \quad x(0) = \eta.$$

It is known that φ is a C^∞ function. Note that $\varphi(t, 0, 0) = 0$ for all $t \in \mathbb{R}$. Find

$$\frac{\partial \varphi}{\partial \eta}(t, 0, 0) \text{ and } \frac{\partial \varphi}{\partial \lambda}(t, 0, 0).$$

17. For each $\lambda \in \mathbb{R}$ denote by $\phi(\cdot, \lambda)$ the maximal solution of the IVP

$$x' = \lambda\, x^2 - t\, x, \quad x(0) = 2.$$

Also, for each $k \in \mathbb{N}^*$ denote by ψ_k the maximal solution of the IVP

$$x' = e^{-k}x^2 - t\, x, \quad x(0) = 2.$$

It is known that φ is a C^∞ function. Find

$$\lim_{k\to\infty} \psi_k(t), \quad \varphi(t, 0) \text{ and } \frac{\partial \varphi}{\partial \lambda}(t, 0).$$

18. For each $\varepsilon \in \mathbb{R}$ denote by $\varphi(\cdot, \varepsilon)$ the maximal solution of the IVP

$$x' = 2t + \varepsilon\, x^2, \quad x(0) = 1.$$

It is known that φ is a C^∞ function. Find $\varphi(t, 0)$, $\frac{\partial \varphi}{\partial \varepsilon}(t, 0)$ and $\frac{\partial^2 \varphi}{\partial \varepsilon^2}(t, 0)$.

19. For each $\eta \in \mathbb{R}$ denote by $\varphi(\cdot, \eta)$ the maximal solution of the IVP

$$x' = \sin(t\, x), \quad x(0) = \eta.$$

Also, for each $k \in \mathbb{N}^*$ denote by ψ_k the maximal solution of the IVP

$$x' = \sin(t\, x), \quad x(0) = \sin\frac{1}{k}.$$

Find

$$\lim_{k\to\infty} \psi_k(t), \quad \varphi(t, 0) \text{ and } \frac{\partial \varphi}{\partial \eta}(t, 0).$$